

“E-Commerce Analytics Project Summary”

Introduction:

This e-commerce analysis examined customer behavior, sales performance, product profitability, marketing effectiveness, customer satisfaction, retention patterns, and customer segmentation using transaction, product, customer, session, and review data from 2020–2024.

Overall, the business demonstrates stable performance with consistent revenue, customer activity, and spending behavior. While acquisition channels successfully generate customers and revenue, retention and customer loyalty remain the largest opportunities for future growth.

Business Questions

Data Quality Assessment

- Are there any missing values, duplicates, or data quality issues?
- Is the dataset suitable for business analysis?
- What preprocessing steps are required before analysis?

Sales Analysis

- How has revenue changed over time?
- What are the monthly and yearly sales trends?
- How many orders are placed each year?
- What is the Average Order Value (AOV)?
- Are there any seasonal sales patterns?

Customer Analysis

- How many customers have made at least one purchase?
- Which countries generate the most customers?
- Which countries generate the highest revenue?
- What percentage of customers are repeat buyers?
- Who are the most valuable customers?

Product Analysis

- Which product categories generate the highest revenue?
- Which products are the best-selling?
- Which products generate the highest profit?
- Are there products with high sales but low profitability?
- How does sales volume differ across product categories?

Marketing Analysis

- Which acquisition channels generate the most revenue?
- Which channels attract the most customers?
- Which channels generate the highest revenue per customer?
- Which devices generate the highest revenue?
- Does customer value vary by acquisition source?

Customer Satisfaction Analysis

- What is the average customer rating?
- How are ratings distributed?
- Which product categories receive the highest ratings?
- Which categories receive the lowest ratings?
- Does customer satisfaction vary by acquisition channel?

Retention & Cohort Analysis

- What percentage of customers return to make another purchase?
- How does retention change over time?
- Which cohorts demonstrate stronger long-term retention?
- When does most customer churn occur?
- Are customer retention patterns improving or declining?

RFM Analysis

- How can customers be segmented based on Recency, Frequency, and Monetary value?
- Which customer segments generate the highest revenue?
- How many customers belong to each segment?
- What proportion of customers are at risk of churn?
- Which segments should be prioritized for retention and loyalty initiatives?

Strategic Recommendations

- What are the main opportunities for revenue growth?
- Which customer segments should receive the most attention?
- How can customer retention be improved?
- Which marketing channels should receive additional investment?
- What actions could increase customer lifetime value (LTV)?

Key Metrics Calculated

Sales Metrics

- Total Revenue – total sales generated during the analyzed period.
- Monthly Revenue – revenue aggregated by month.
- Yearly Revenue – revenue aggregated by year.
- Number of Orders – total completed orders.
- Average Order Value (AOV) – average revenue generated per order.
- Revenue by Category – total revenue generated by each product category.
- Revenue by Product – total revenue generated by individual products.
- Profit – revenue minus product cost.
- Profit Margin (%) – profit as a percentage of revenue.

Customer Metrics

- Unique Customers – number of customers who made at least one purchase.

- Repeat Customer Rate – percentage of customers who placed more than one order.
- Revenue per Customer – average revenue generated per customer.
- Top Customer Revenue – spending generated by the highest-value customers.
- Customer Distribution by Country – number of customers by geographic location.

Product Metrics

- Units Sold – total quantity of products sold.
- Top-Selling Products – products with the highest sales volume.
- Top Revenue Products – products generating the highest revenue.
- Top Profit Products – products generating the highest profit.
- Category Profitability – profit performance across product categories.

Marketing Metrics

- Revenue by Acquisition Channel – revenue generated by each traffic source.
- Customers by Acquisition Channel – number of customers acquired through each source.
- Revenue per Customer by Channel – average customer value by acquisition source.
- Revenue by Device Type – sales generated by desktop, mobile, and tablet users.
- Average Order Value by Device – comparison of spending behavior across devices.

Customer Satisfaction Metrics

- Average Rating – overall customer satisfaction score.
- Rating Distribution – number of reviews by rating level.
- Average Rating by Category – customer satisfaction across product categories.
- Average Rating by Acquisition Source – customer satisfaction by traffic channel.
- Review Volume – total number of customer reviews.

Retention Metrics

- Repeat Customer Rate – percentage of returning customers.
- Monthly Retention Rate – customer retention over time.
- Cohort Retention – retention performance by acquisition cohort.
- Customer Churn Pattern – decline in customer activity after first purchase.

RFM Metrics

- Recency – days since the customer's last purchase.
- Frequency – number of purchases made by a customer.
- Monetary Value – total revenue generated by a customer.
- RFM Score – combined customer value score.
- Customer Segment Distribution – share of customers in each RFM segment.
- Revenue by Segment – revenue contribution of each customer segment.

Derived Variables

To support the analysis, several new variables were created from the original dataset.

Date-Based Variables

- order_year – year extracted from the order date.
- order_month – month extracted from the order date.
- year_month – combined year-month field used for trend analysis.
- days_since_last_purchase – number of days since the customer's most recent purchase.

Revenue & Profit Variables

- line_revenue – revenue generated by each order item.

quantity × sale_price

- profit

sale_price - cost

- profit_margin (%)

$(\text{profit} / \text{sale_price}) \times 100$

- customer_revenue – total revenue generated by a customer.
- category_revenue – total revenue generated by a product category.

Customer Variables

- order_count – total number of orders placed by a customer.
- repeat_customer_flag – indicates whether a customer placed more than one order.

Retention Variables

- first_purchase_date – customer's first order date.
- cohort_month – month of first purchase.
- months_since_first_purchase – number of months between the first purchase and subsequent orders.
- retention_rate (%) – percentage of customers retained over time.

Customer Satisfaction Variables

- average_rating – average review score.
- rating_category
 - Low (1–2)
 - Medium (3)
 - High (4–5)

Marketing Variables

- revenue_per_customer

$\text{total_revenue} / \text{unique_customers}$

- channel_revenue_share (%) – contribution of each acquisition source to total revenue.
- device_revenue_share (%) – contribution of each device type to total revenue.

RFM Variables

- Recency

max_order_date - customer's last purchase date

- Frequency

number of purchases per customer

- Monetary

total spending per customer

- R Score – Recency score (1–5).
- F Score – Frequency score (1–5).
- M Score – Monetary score (1–5).
- RFM Score – combined RFM value.
- Customer Segment
 - Best Customers
 - Loyal Customers
 - At Risk Customers
 - Lost Customers
 - Others

The project includes:

- Data cleaning and preprocessing in Python,
- SQL-based analytical queries,
- Exploratory data analysis (EDA),
- Cohort and retention analysis,
- RFM customer segmentation,
- Interactive dashboard in Power BI which consists of 4 pages(Executive overview,customer analysis,product performance and funnel analysis).

Dataset description:

The Dataset used for this project spans over a period of 2020 and 2025 years.

There are tables such as :

- customers.csv — customer profiles, signup dates, country, opt-in
- products.csv — catalog with categories, prices, costs, margins
- sessions.csv — session metadata (device, source, start time, country)
- events.csv — page_view / add_to_cart / checkout / purchase with timestamps
- orders.csv — order headers (payment, discount, totals)
- order_items.csv — line items (quantity, unit price, line total)
- reviews.csv — product ratings & short text reviews

Main preprocessing steps included:

- handling missing values,
- checking duplicates,
- converting date columns to datetime format,
- detecting negative and abnormal values

```
for name, df in {  
    'customers': customers,  
    'products': products,  
    'sessions': sessions,  
    'events': events,  
    'orders': orders,  
    'order_items': order_items,  
    'reviews': reviews  
}.items():  
  
    print(f'\n{name.upper()}')  
    print('Duplicates:', df.duplicated().sum())  
    print('\nMissing values:')  
    print(df.isna().sum())
```

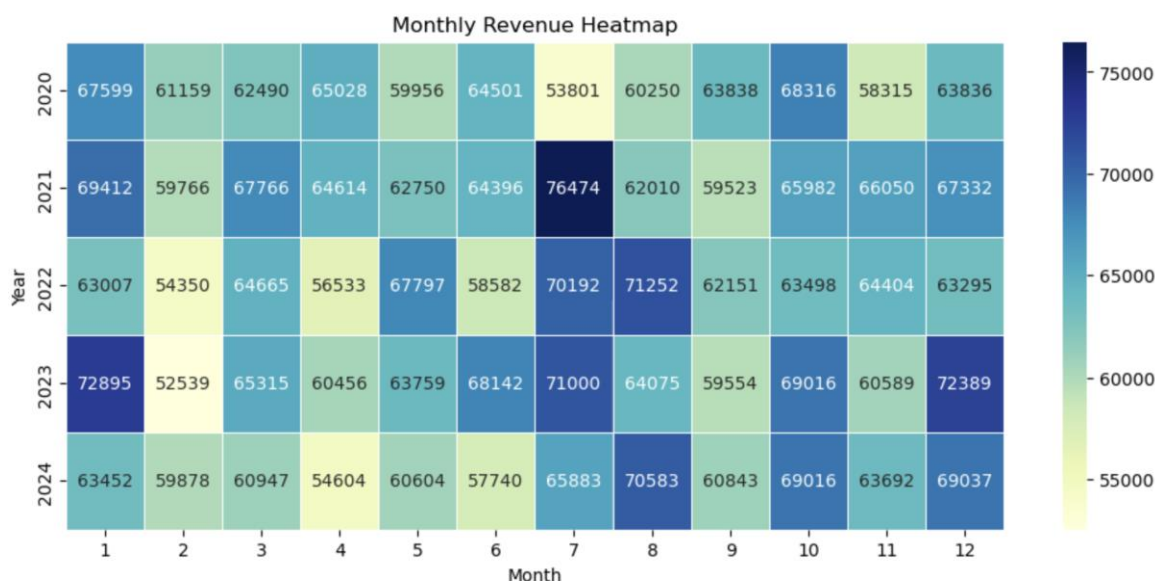
```
customers['signup_date'] = pd.to_datetime(customers['signup_date'])  
  
sessions['start_time'] = pd.to_datetime(sessions['start_time'])  
  
events['timestamp'] = pd.to_datetime(events['timestamp'])  
  
orders['order_time'] = pd.to_datetime(orders['order_time'])  
  
reviews['review_time'] = pd.to_datetime(reviews['review_time'])
```

Key Business Findings

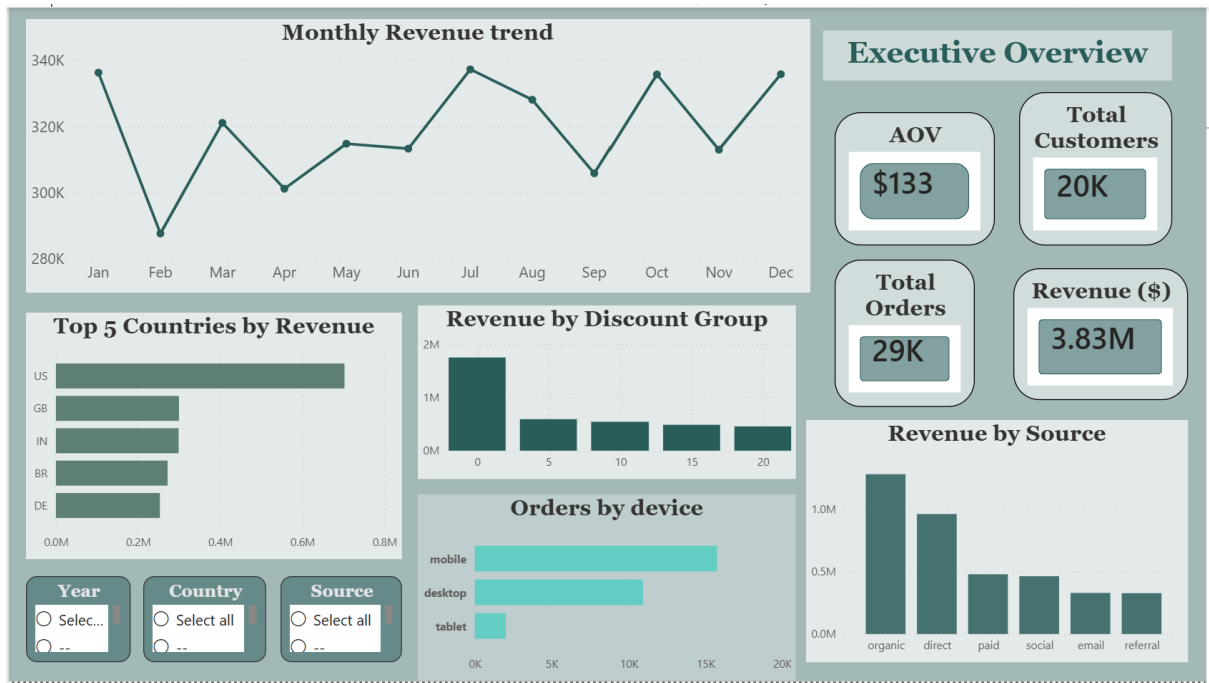
1. Data Quality

- The dataset was generally clean and suitable for analysis.
- No significant missing values were found in core business tables.
- 73 duplicate records were identified and removed from the order_items table.
- Date fields were standardized for time-series analysis.
- Only complete years (2020–2024) were retained to ensure accurate year-over-year comparisons.

2. Sales Performance



- Annual revenue remained stable between **\$749K and \$780K** throughout the analyzed period.
- Order volume remained consistent at approximately **5.7K orders per year**.
- Average Order Value (AOV) remained stable between **\$130 and \$136**.
- The customer base remained stable at roughly **5,000 customers annually**.
- No significant long-term growth or decline trends were observed.
- Revenue fluctuations were primarily driven by order volume rather than changes in customer spending.



Business Insight

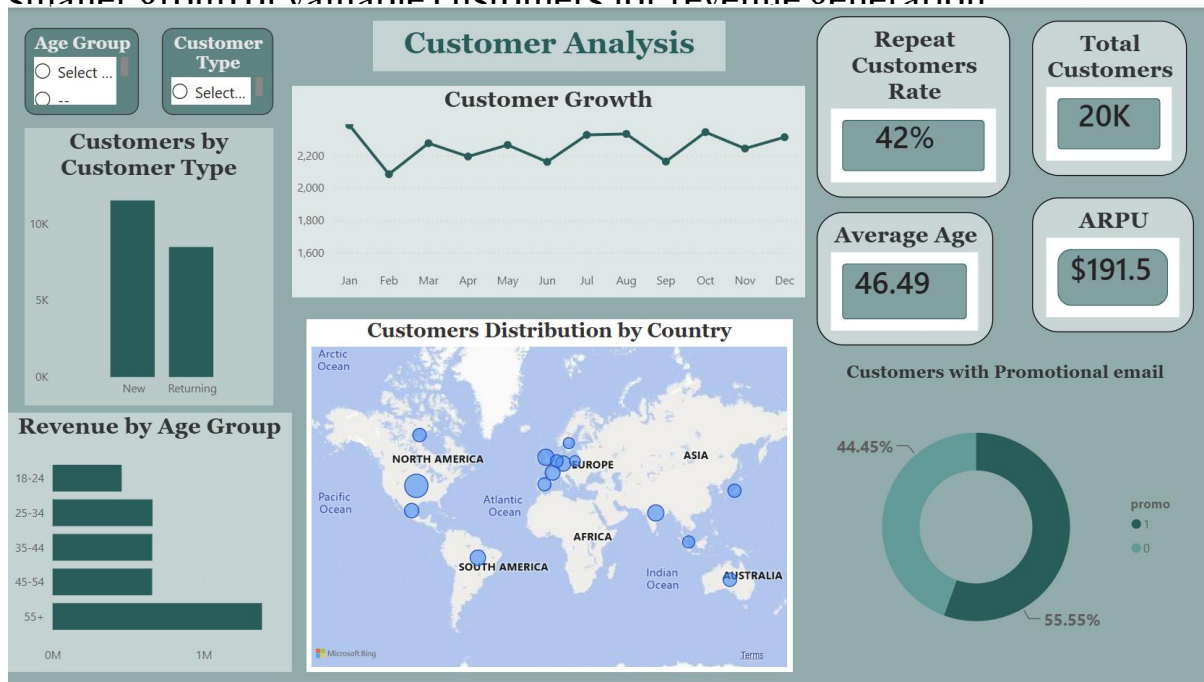
The company appears to be a mature business with predictable demand and purchasing behavior but limited organic growth.

3. Customer Analysis

- **15,260 unique customers** completed at least one purchase.
- The United States represented the largest customer base and generated the highest revenue.
- Approximately **76.3% of registered customers completed a purchase**.
- Customer revenue is concentrated among a relatively small number of high-value buyers.
- Top customers generated between **\$2.1K and \$3.0K** in total revenue.

Business Insight

The business has strong customer acquisition but relies heavily on a smaller group of valuable customers for revenue generation

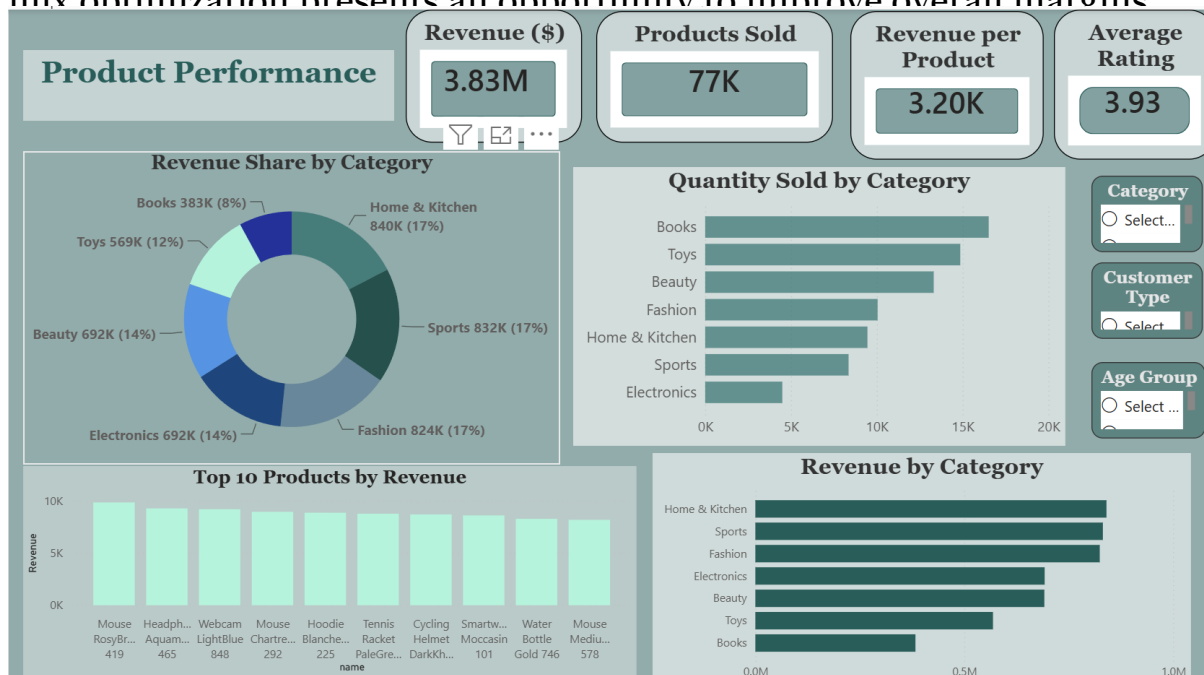


4. Product Analysis

- **Home & Kitchen** generated the highest revenue, followed closely by Sports and Fashion.
- **Books** was the best-selling category by units sold but generated the lowest revenue.
- Electronics generated strong revenue despite low sales volume, indicating higher average selling prices.
- Profit margins varied significantly across products, ranging from approximately **22% to 50%**.
- Several products generated strong revenue while maintaining relatively low profit margins.

Business Insight

Sales volume does not necessarily translate into profitability. Product mix optimization presents an opportunity to improve overall margins



5. Marketing Analysis

- **Organic traffic** was the strongest acquisition channel.
- Organic customers generated:
 - The highest revenue
 - The largest customer base
 - The highest revenue per customer
- Direct traffic performed similarly well and demonstrated strong brand loyalty.
- Paid traffic successfully acquired customers but produced lower customer value.
- Email marketing generated the lowest revenue per customer.
- Customer value was relatively consistent across device types.

Business Insight

Marketing investment should prioritize Organic and Direct channels while improving targeting efficiency in Paid and Email campaigns.

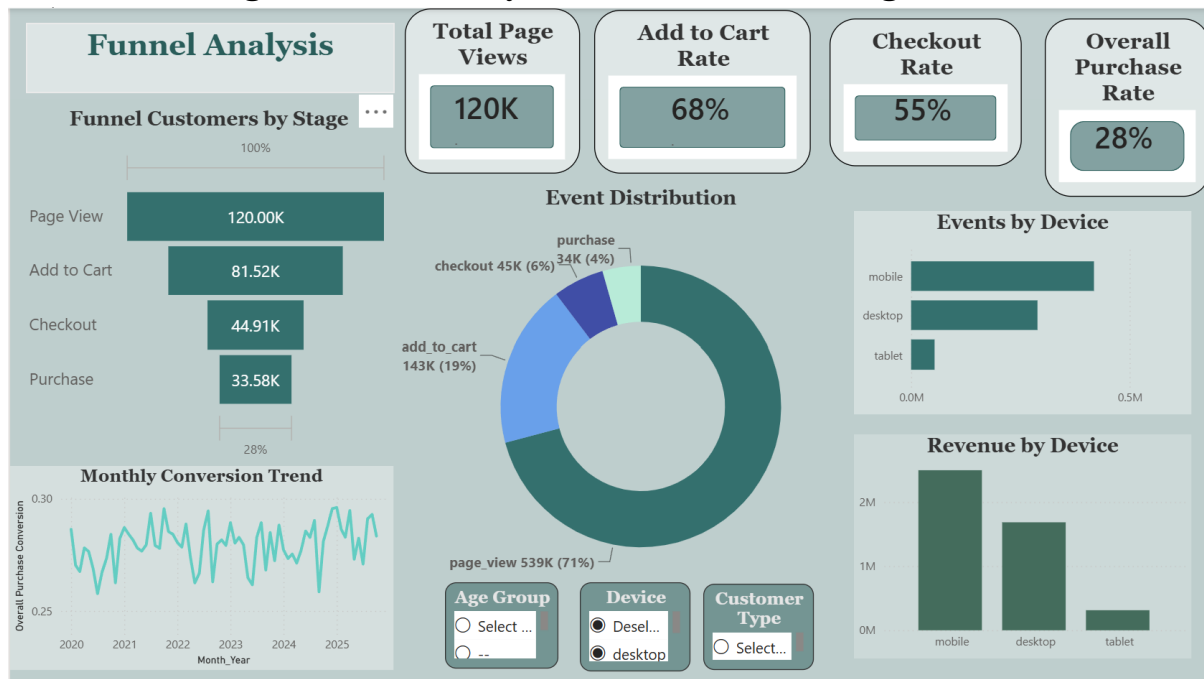
6. Customer Satisfaction Analysis

- Average rating reached **3.93/5**.

- More than **75% of reviews were rated 4 or 5 stars**.
- Beauty products achieved the highest customer satisfaction scores.
- Electronics received the lowest ratings among product categories.
- Direct traffic customers reported the highest satisfaction levels.
- Email-acquired customers reported slightly lower ratings.

Business Insight

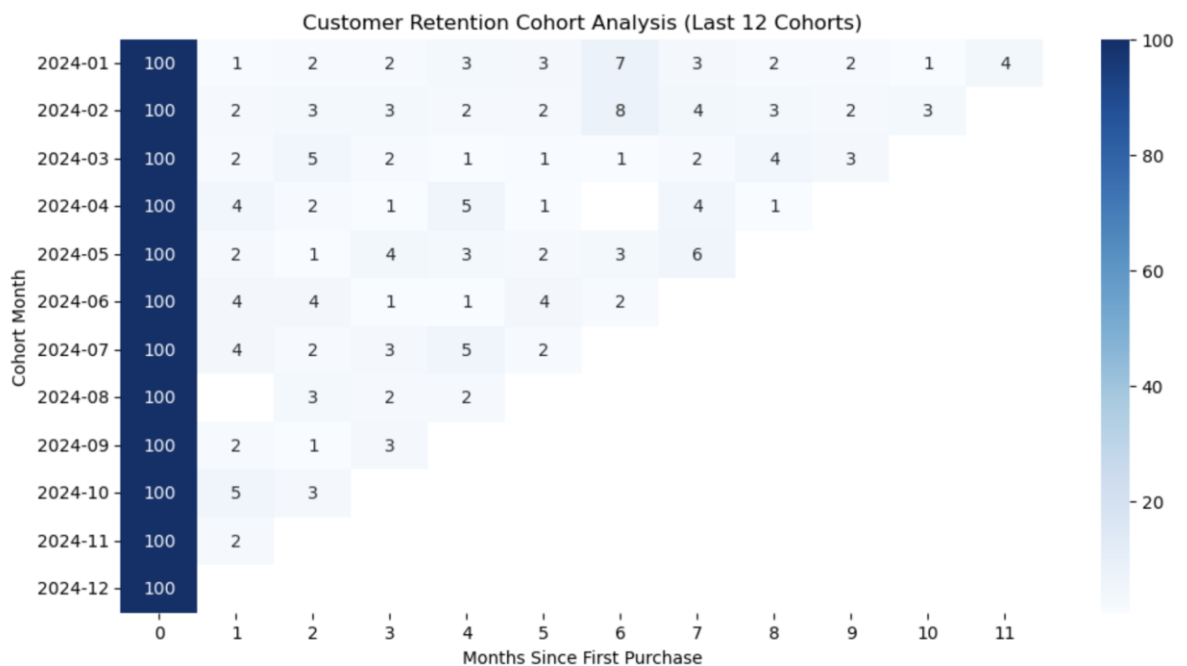
Overall customer satisfaction is strong, but Electronics products require further investigation to identify drivers of lower ratings.



7. Retention & Cohort Analysis

- Overall repeat customer rate reached **55.54%**.
- Monthly retention remained relatively low, generally between **1% and 8%** after the acquisition month.
- Customers frequently return to make additional purchases, but not on a predictable monthly schedule.
- Cohort behavior remained relatively consistent over time.

- Most customer churn occurs shortly after the first purchase.



Business Insight

The company has a solid base of repeat buyers but lacks strong recurring purchasing behavior.

8. RFM Segmentation

Customer Distribution

- Others: **32%**
- At Risk Customers: **32%**
- Loyal Customers: **22%**
- Lost Customers: **8%**
- Best Customers: **6%**

Revenue Contribution

- At Risk Customers generated the largest share of revenue.
- Best Customers represented only 6% of customers but contributed substantial revenue.
- Loyal Customers provided a stable stream of repeat purchases.

Business Insight

A significant portion of revenue comes from customers who may be at risk of churn, making retention efforts critically important.

Strategic Recommendations

Customer Retention

- Implement loyalty and rewards programs.
- Launch personalized re-engagement campaigns for At Risk customers.
- Encourage second purchases within 30–60 days of the initial order.

Marketing Optimization

- Increase investment in Organic acquisition.
- Improve Paid campaign targeting and ROI measurement.
- Personalize Email marketing campaigns.

Product Strategy

- Promote high-margin products more aggressively.
- Review low-margin, high-volume products for pricing opportunities.
- Investigate lower customer satisfaction in Electronics.

Customer Experience

- Analyze negative reviews to identify recurring issues.
- Replicate successful practices from top-rated categories such as Beauty.

Revenue Growth

- Focus on converting Loyal Customers into Best Customers.
- Prioritize retention of high-value segments before investing heavily in new customer acquisition.

Final Conclusion

The business demonstrates strong operational stability, healthy customer satisfaction, and effective customer acquisition. However, future growth will depend less on acquiring additional customers and more on improving retention, increasing customer lifetime value, and strengthening relationships with high-value customer segments. By focusing on loyalty initiatives, customer experience improvements, and marketing optimization, the company can unlock significant long-term revenue growth without substantially increasing acquisition costs.